Math-4141

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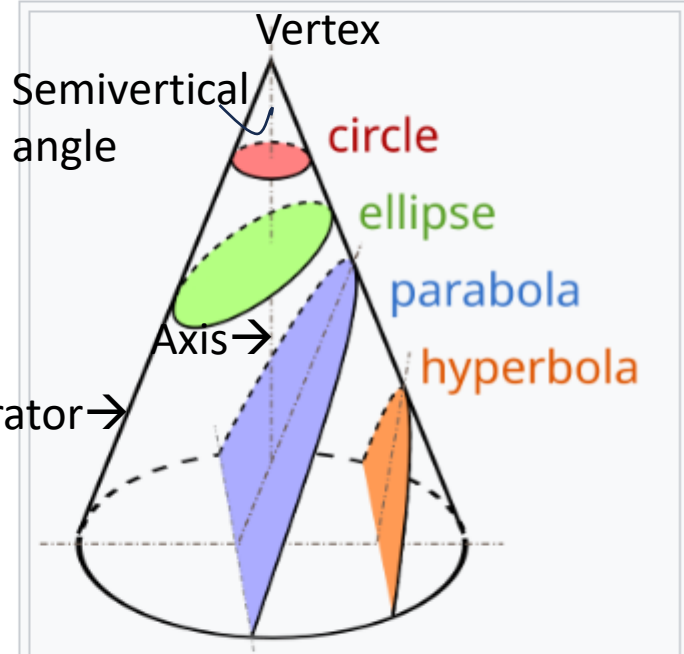
General Equation of Second Degree

Conic section

From Wikipedia, the free encyclopedia

A **conic section**, **conic** or a **quadratic curve** is a curve obtained from a cone's surface intersecting a plane. The three types of conic section are the hyperbola, the parabola, and the ellipse; the circle is a special case of the ellipse, though it was sometimes called as a fourth type. The ancient Greek mathematicians studied conic sections, culminating around 200 BC with Apollonius of Perga's systematic work on their properties.

The conic sections in the Euclidean plane have various distinguishing properties, many of which can be used as alternative definitions. One such property defines a non-circular conic[1] to be the set of those points whose distances to some particular point, called a *focus*, and some particular line, called a *directrix*, are in a fixed ratio, called the *eccentricity*. The type of conic is determined by the value of the eccentricity. In analytic geometry, a conic may be defined as a plane algebraic curve of degree 2; that is, as the set of points whose coordinates satisfy a quadratic equation in two variables which can be written in the form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. The geometric properties of the conic can be deduced from its equation.



The black boundaries of the colored regions are conic sections. Not shown is the other half of the hyperbola, which is on the unshown other half of the double cone.

Eccentricity, focus and directrix

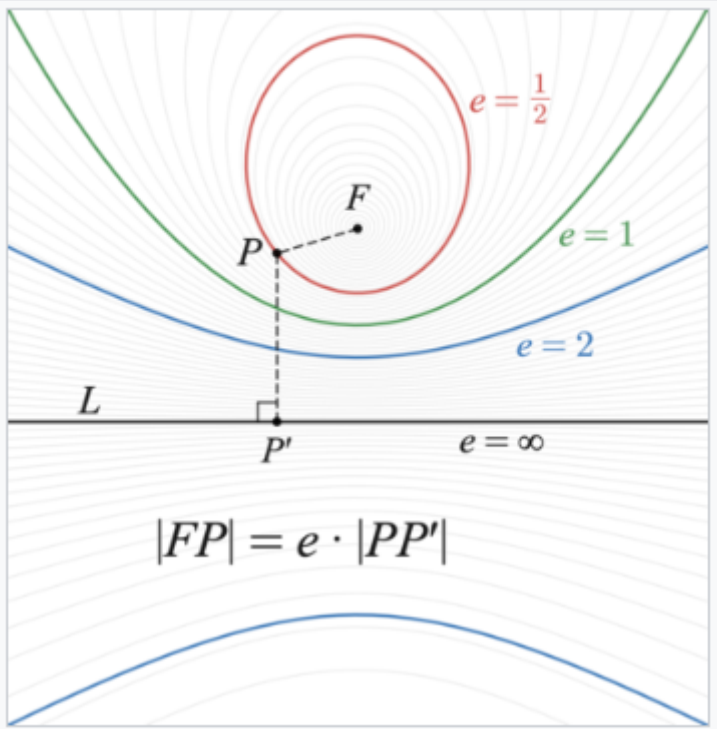
Alternatively, one can define a conic section purely in terms of plane geometry: it is the locus of all points P whose distance to a fixed point F (called the *focus*) is a constant multiple e (called the *eccentricity*) of the distance from P to a fixed line L (called the *directrix*). For $0 < e < 1$ we obtain an ellipse, for $e = 1$ a parabola, and for $e > 1$ a hyperbola.

A circle is a limiting case and is not defined by a focus and directrix in the Euclidean plane. The eccentricity of a circle is defined to be zero and its focus is the center of the circle, but its directrix can only be taken as the line at infinity in the projective plane.[2]

Alternatively, an ellipse can be defined in terms of two focus points, as the locus of points for which the sum of the distances to the two foci is $2a$; while a hyperbola is the locus for which the difference of distances is $2a$. (Here a is the semi-major axis defined below.) A parabola may also be defined in terms of its focus and latus rectum line (parallel to the directrix and passing through the focus): it is the locus of points whose distance to the focus plus or minus the distance to the line is equal to $2a$; plus if the point is between the directrix and the latus rectum, minus otherwise.

- $e = 0 \rightarrow$ Infinitesimal circle with centre at Focus.
- $e \rightarrow \infty$: Infinitesimally seperateed pair of straight lines.
- Infinitely distant directrix➔ A circle with finite radius.
- Infinitely distant focus➔ A pair of lines of finite separation.

Conic Section - A Geometric Construction using Eccentricity



Conic sections of varying eccentricity sharing a focus point and directrix line, including an ellipse (red, $e = 1/2$), a parabola (green, $e = 1$), and a hyperbola (blue, $e = 2$). The conic of eccentricity 0 in this figure is an infinitesimal circle centered at the focus, and the conic of eccentricity ∞ is an infinitesimally separated pair of lines.

A circle of finite radius has an infinitely distant directrix, while a pair of lines of finite separation have an infinitely distant focus.

Conic parameters

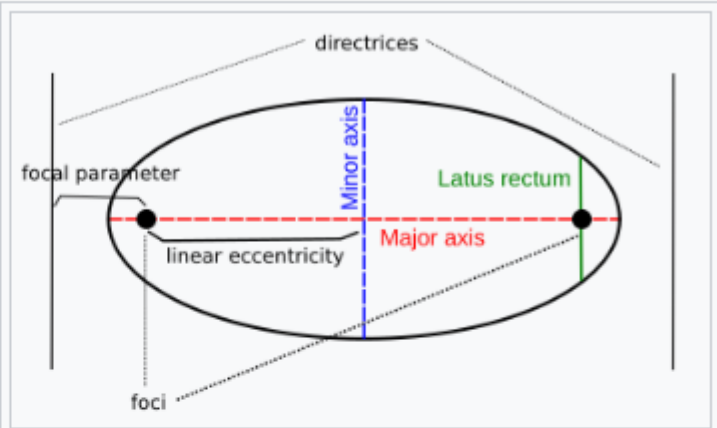
In addition to the eccentricity (e), foci, and directrix, various geometric features and lengths are associated with a conic section.

The *principal axis* is the line joining the foci of an ellipse or hyperbola, and its midpoint is the curve's *center*. A parabola has no center.

The *linear eccentricity* (c) is the distance between the center and a focus.

The *latus rectum* is the chord parallel to the directrix and passing through a focus; its half-length is the *semi-latus rectum* (ℓ).

The *focal parameter* (p) is the distance from a focus to the corresponding directrix.



Conic parameters in the case of an ellipse

The *major axis* is the chord between the two vertices: the longest chord of an ellipse, the shortest chord between the branches of a hyperbola. Its half-length is the *semi-major axis* (a). When an ellipse or hyperbola are in standard position as in the equations below, with foci on the x-axis and center at the origin, the vertices of the conic have coordinates $(-a, 0)$ and $(a, 0)$, with a non-negative.

The *minor axis* is the shortest diameter of an ellipse, and its half-length is the *semi-minor axis* (b), the same value b as in the standard equation below. By analogy, for a hyperbola the parameter b in the standard equation is also called the semi-minor axis.

The following relations hold:[6]

- $\ell = pe$
- $c = ae$
- $p + c = \dfrac{a}{e}$

For conics in standard position, these parameters have the following values, taking $a, b > 0$.

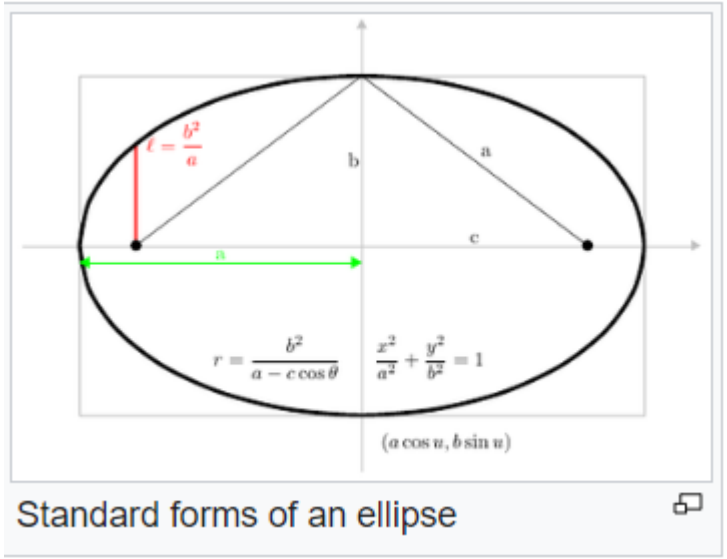
conic section	equation	eccentricity (e)	linear eccentricity (c)	semi-latus rectum (ℓ)	focal parameter (p)
circle	$x^2 + y^2 = a^2$	0	0	a	∞
ellipse	$\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$	$\sqrt{1 - \dfrac{b^2}{a^2}}$	$\sqrt{a^2 - b^2}$	$\dfrac{b^2}{a}$	$\dfrac{b^2}{\sqrt{a^2 - b^2}}$
parabola	$y^2 = 4ax$	1	N/A	$2a$	$2a$
hyperbola	$\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$	$\sqrt{1 + \dfrac{b^2}{a^2}}$	$\sqrt{a^2 + b^2}$	$\dfrac{b^2}{a}$	$\dfrac{b^2}{\sqrt{a^2 + b^2}}$

Standard forms in Cartesian coordinates

After introducing Cartesian coordinates, the focus-directrix property can be used to produce the equations satisfied by the points of the conic section.[7] By means of a change of coordinates (rotation and translation of axes) these equations can be put into *standard forms*.[8] For ellipses and hyperbolas a standard form has the x-axis as principal axis and the origin (0,0) as center. The vertices are $(\pm a, 0)$ and the foci $(\pm c, 0)$. Define b by the equations $c^2 = a^2 - b^2$ for an ellipse and $c^2 = a^2 + b^2$ for a hyperbola. For a circle, $c = 0$ so $a^2 = b^2$, with *radius* $r = a = b$. For the parabola, the standard form has the focus on the x-axis at the point $(a, 0)$ and the directrix the line with equation $x = -a$. In standard form the parabola will always pass through the origin.

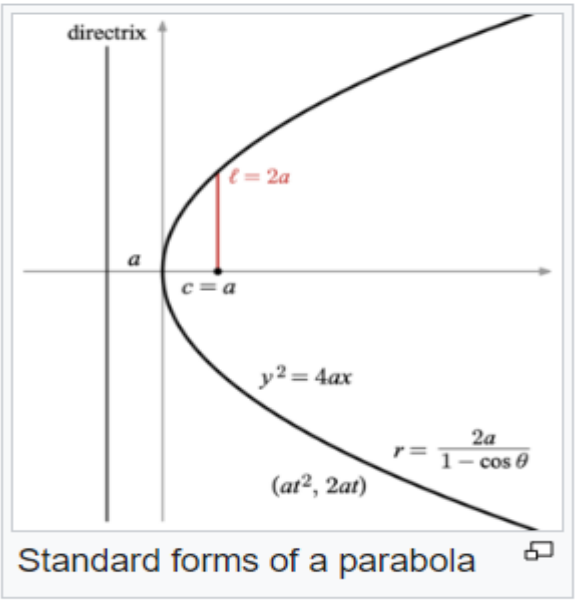
For a *rectangular* or *equilateral* hyperbola, one whose asymptotes are perpendicular, there is an alternative standard form in which the asymptotes are the coordinate axes and the line $x = y$ is the principal axis. The foci then have coordinates (c, c) and $(-c, -c)$.[9]

- Circle:
$$x^2 + y^2 = a^2,$$
- Ellipse:
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$
- Parabola:
$$y^2 = 4ax, \text{ with } a > 0,$$
- Hyperbola:
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$
- Rectangular hyperbola:[10]
$$xy = c^2.$$



Standard forms of an ellipse

Ellipse Plotter



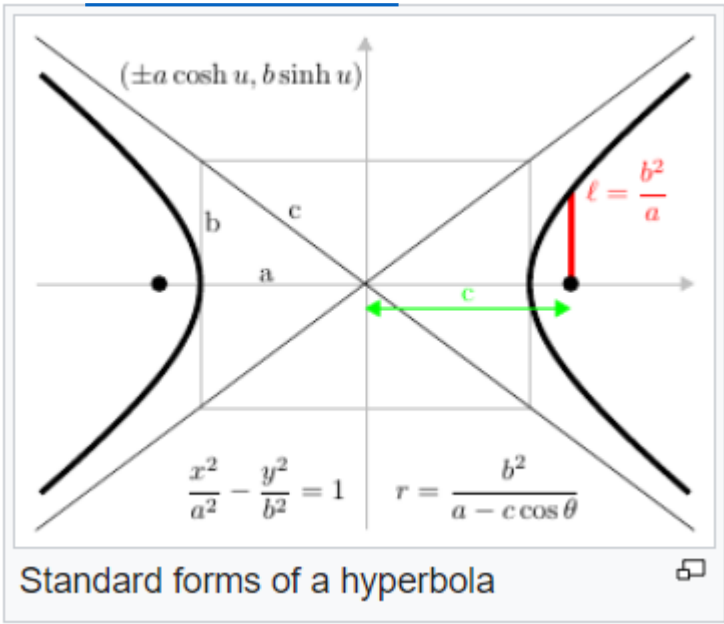
Standard forms of a parabola

Parabola Plotter

The first four of these forms are symmetric about both the x-axis and y-axis (for the circle, ellipse and hyperbola), or about the x-axis only (for the parabola). The rectangular hyperbola, however, is instead symmetric about the lines $y = x$ and $y = -x$.

These standard forms can be written parametrically as,

- Circle:
$$(a\cos\theta, a\sin\theta),$$
- Ellipse:
$$(a\cos\theta, b\sin\theta),$$
- Parabola:
$$(at^2, 2at),$$
- Hyperbola:
$$(a\sec\theta, b\tan\theta), \text{ or } (\pm a\cosh\psi, b\sinh\psi),$$
- Rectangular hyperbola:
$$\left(dt, \frac{d}{t}\right), \text{ where } d = \frac{c}{\sqrt{2}}.$$



Standard forms of a hyperbola

Hyperbola Plotter

General Cartesian form

In the Cartesian coordinate system, the graph of a quadratic equation in two variables is always a conic section (though it may be degenerate),[a] and all conic sections arise in this way. The most general equation is of the form[11]

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

with all coefficients real numbers and A, B, C not all zero.

Discriminant

The conic sections described by this equation can be classified in terms of the value $B^2 - 4AC$, called the discriminant of the equation.[13] Thus, the discriminant is -4Δ where Δ is the matrix determinant $\begin{vmatrix} A & B/2 \\ B/2 & C \end{vmatrix}$.

If the conic is non-degenerate, then:[14]

- if $B^2 - 4AC < 0$, the equation represents an ellipse;
  - if $A = C$ and $B = 0$, the equation represents a circle, which is a special case of an ellipse;
- if $B^2 - 4AC = 0$, the equation represents a parabola;
- if $B^2 - 4AC > 0$, the equation represents a hyperbola;
  - if $A + C = 0$, the equation represents a rectangular hyperbola.

In the notation used here, A and B are polynomial coefficients, in contrast to some sources that denote the semimajor and semiminor axes as A and B.

Invariants

The discriminant $B^2 - 4AC$ of the conic section's quadratic equation (or equivalently the determinant $AC - B^2/4$ of the 2 × 2 matrix) and the quantity $A + C$ (the trace of the 2 × 2 matrix) are invariant under arbitrary rotations and translations of the coordinate axes,[14][15][16] as is the determinant of the 3 × 3 matrix above.[17]:pp. 60–62 The constant term F and the sum $D^2 + E^2$ are invariant under rotation only.[17]:pp. 60–62

Polar coordinates

In polar coordinates, a conic section with one focus at the origin and, if any, the other at a negative value (for an ellipse) or a positive value (for a hyperbola) on the x-axis, is given by the equation

$$r = \frac{l}{1 + e\cos\theta},$$

where e is the eccentricity and l is the semi-latus rectum.

As above, for $e = 0$, the graph is a circle, for $0 < e < 1$ the graph is an ellipse, for $e = 1$ a parabola, and for $e > 1$ a hyperbola.

The polar form of the equation of a conic is often used in dynamics; for instance, determining the orbits of objects revolving about the Sun.[20]

- **Solve all examples from the book:**
  Co-ordinate Geometry with Vector Analysis
  - Rahman & Bhattacharjee.

Art. 43. Conic Section regarded as locus.

Definition : If a point P moves in a plane such a way that the ratio of its distance PS from a fixed point S in the plane to its perpendicular distance PM from a fixed straight line XM in it, is always a constant, the locus of the point P is called a Conic Section or briefly a **Conic.**

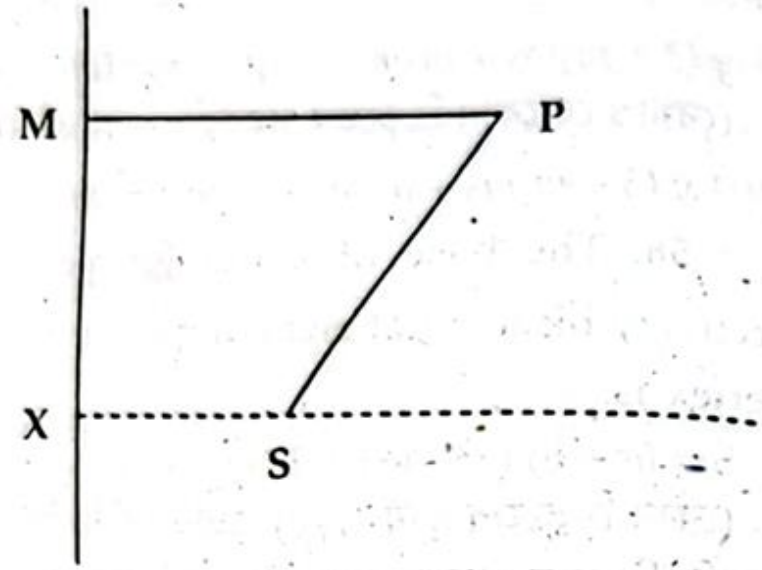


Fig. 16

The constant ratio is called the **eccentricity** of the conic and is generally represented by the letter '*e*'. The fixed point S is called the **focus and** the fixed straight line XM is called **directrix** of the conic.

The conic is called a **Parabola, and Ellipse or a Hyperbola according** as the eccentricity $e =$, $<$ or $>$. 1.

That is for a **Parabola,** $SP = PM$

for an **Ellipse,** $SP = ePM \quad e < 1$

for **Hyperbola,** $SP = ePM \quad e > 1$

The eccentricity is a positive number.

Art. 45. Cartesian equation of a conic given its focus, directrix and eccentricity.

Let S (α, β) be the given focus and $lx + my + n = 0$ the given directrix of a conic whose eccentricity is e. Then if P (x, y) be any position of the moving point tracing the conic, wd have from fig. 18.

$$SP^2 = (x - \alpha)^2 + (y - \beta)^2 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

$$\text{and} \quad PM = \frac{lx + my + n}{(\sqrt{l^2 + m^2})} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

$\therefore$ $SP^2 = ePM$ *by* Art. 43.

$$SP = e^2 PM^2$$

$$\text{or}, (x - \alpha)^2 + (y - \beta)^2 = e^2 \frac{(lx + my + n)^2}{l^2 + m^2} \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

$$\text{or}, x^2 \{(l^2 + m^2) - e^2 l^2\} - 2e^2 lmxy + y^2 \{(l^2 + m^2) - e^2 m\}$$
$$- 2x \{\alpha (l^2 + m^2) - e^2 nl^2\} - 2 \{\beta(l^2 + m^2) - e^2 mn\}$$
$$+ (l^2 + m^2)(\alpha^2 + \beta^2) - e^2 n^2 = 0$$

which is a relation connecting the co-ordinates (x, y) of their moving point P in any position. This equation of the conic is traced out by the point P.

When $e < 1$, the equation represents an ellipse, when $e > 1$, it represents a hyperbola and a parabola when $e = 1$.

The equation (4) is of the form.

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

which is called the General Equation of the Second degree in a plane. If the focus is chosen for the origin, $\alpha = 0; \beta = 0$, so that the equation (3) is

$$(l^2 + m^2)(x^2 + y^2) = e^2 (lx + my + n)^2 \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

With rectangular axis, it represents a conic; whose focus is at the origin whose eccentricity is $\sqrt{(l^2 + m^2)}$

Art. 48. From the above discussions the general equation of the second degree.

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent

(1) a pair of straight lines if the determinant

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

(i) two parallel lines if $\Delta = 0, ab = h^2$

(ii) two perpendicular lines if $\Delta = 0, a + b = 0$

(2) a circle if $a = b, h = 0,$

(3) a parabola if $ab = h^2, \Delta \neq 0$

(4) an ellipse if $ab - h^2 > 0, \Delta \neq 0$

(5) a hyperbola if $ab - h^2 < 0, \Delta \neq 0$

(6) a rectangular hyperbola if. $a + b = 0, ab - h^2 < 0, \Delta \neq 0$

Note : $\Delta = abc + 2fgh - af^2 - bg^2 - cg^2 =$

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Art. 49. Centre of a conic.

Definition : The centre of a conic is the point in the plane of the conic such that every chord of the conic passing through it is bisected at it.

Let the equation to the conic be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

and let $\frac{x - x_1}{l} = \frac{y - y_1}{m} = r$ ... ... (3)

be the equation of any line through (x_1, y_1)

If the live (2) meets the conic (1) at a distance r from (x_1, y_1) then $(x = x_1 + lr, y = y_1 + mr)$ is oint of intersection of (1) and (2) The from (1)

$$a(x_1 + lr)^2 + 2h(x_1 + lr)(y_1 + mr) + b(y_1 + mr)^2 + 2g(x_1 + lr) + 2f(y_1 + mr) + c = 0$$

or, $(al^2 + 2hlm + bm^2) r^2 + 2r \{ax_1 + hy_1 + g) l + (hx_1 + by_1 + f) m\} + (ax_1^2 + 2hx_1y_1 + by_1^2 + {}_1 + 2fy_1 + c) = 0$ ... ... (4)

From Eq. (4) we see that r has two roots say r_1 and r_2 which are the distances from (x_1, y_1) the line (2) inersects the conic (1) at two points.

If (x_1, y_1) be the middle point of the chord, then r_1 and r_2 have equal but opposite sign i. *e.* $= -r_2$, or $r_1 + r_2 = 0$

The sum of the roots of r in (3) $= 0$

or $(ax_1 + by_1 + g) l + (hx_1 + by_1 + f) m = 0$ ... ... (5)

If (x_1, y_1) is the centre of the conic, then the relation (4) holds good for all values of l and m

$\therefore \quad ax_1 + hy_1 + g = 0$ ... ... (6)

$hx_1 + hy_1 + f = 0$ ... ... (7)

The above two equations given the centre of the conic (1). Solve (6) and (7) for x_1 and y_1

$$\therefore \quad x_1 = \frac{hf - bg}{ab - h^2}, \quad y_1 = \frac{gh - af}{ab - h^2}$$

Hence the co-ordinates of centre (1) is $\left(\frac{hf - bg}{ab - h^2}\right), \left(\frac{gh - af}{ab - h^2}\right)$

Cor. 1. When $ab - h^2 = 0$, the co-ordinates of the centre are at infinity and the curve is a parabola which has no centre.

Cor. 2. If $hf - bg = 0$ and $ab - h^2 = 0$

i. e. $a/h = h/b = g/f$

The equations (3) and (4) represent the same line. Is this case any point on the straight line (3) or (4) may be regarded as the centre, for every such point satisfies the definition of the centre.

The locus in this case is a pair of parallel straight lines.

Art. 51. Axes : To find the lengths and position of the axes of the conic. [C. U. 1986]

$ax^2 + 2hxy + by^2 = 1$ ... ... (1)

Let the axes be rectangular.

Let the conic be cut by a concentric circle of radius r. The diameters through the points of intersection will be equally inclined to the axes of the conic, and will be coincident if the radius of the circle be equal to either of the semi-axes of the conic.

Now suppose that equation of the circle as $\frac{x^2 + y^2}{r^2} = 1$ ... (2)

Making (1) homogeneous with the help of (2), we have

$$\left(a - \frac{1}{r^2}\right)x^2 + 2hxy + \left(b - \frac{1}{r^2}\right)y^2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

which represents a pair of straight lines passing through the origin and the points of intersection of (1) and (2). When these lines (diameters) coincide they become axes of conic (1).

They will be coincident (by condition of equal roots)

$$(a - 1/r^2)(b - 1/r^2) = h^2 \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

$$\text{or, } \frac{1}{r^4} - (a+b)\frac{1}{r^2} + ab - h^2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

This is a quadratic equation in r^2. Hence the square of lengths of the semi-axes of the conic. (4) are the roots of (5)

Let r_1^2 and r_2^2 be the two roots. The semi-axes of the conic are of lengths r_1 and r_2.

Now multiply the equation (3) by $(a - 1/r^2)$ then it becomes

$$(a - 1/r^2)^2 x^2 + 2h(a - 1/r^2)xy + (a - 1/r^2)(b - 1/r^2)y^2 = 0$$

$$\text{or, } (a - 1/r^2)^2 x^2 + 2h(a - 1/r^2)xy + h^2y^2 = 0 \text{ by (4)}$$

$$\text{or, } \{(a - 1/r^2)x + hy\}^2 = 0 \text{ or, } (a - 1/r^2)x + hy = 0$$

The equations of the axes are therefore

$$\left(a - \frac{1}{r_1^2}\right)x + hy = 0 \text{ and } \left(a - \frac{1}{r_2^2}\right)x + hy = 0 \quad \ldots \quad \ldots \quad (6)$$

and length of the corres ponding semi-axes are r_1 and r_2 where r_1^2 and r_2^2 are the roots of (5)

Cor. For the general equation.

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ... ... (7)

which reduces to the equation

$ax^2 + 2hxy + by^2 = k$ by Eq 8 Art. 50.

where $k = -c_1$ from (7) Art. 50

Making the equation homogeneous by (2) we have

$$ax^2 + 2hxy + by^2 = k\frac{x^2 + y^2}{r^2}$$

$$\text{or, } \left(a - \frac{k}{r^2}\right)x^2 + 2hxy + \left(b - \frac{k}{r^2}\right)y^2 = 0 \quad \text{.... ... ... (8)}$$

In this case the square of the Semi-axes are given by (sec relation 4)

$(a - k/r^2)(b - k/r^2) = h^2$ ... ... (9)

and the equations of the axes are given by

$(a - k/r_1^2)x + hy = 0$ and $(a - k/r_2^2)\ x + hy = 0$ ... ... (10)

Note : As the minor or conjugate axis is perpendicular to the major or transverse axis, the equation of minor or conjugate axis is easily obtained if equation of the major or transverse axis is known. say if $5x + 3y + 6 = 0$ is the equation of major axis the equation of the minor axis is $3x - 5y + k = 0$, Since it pass through the centre, then k is known.

Sometimes the minor or conjugate axis obtained from the fromula $(b - 1/r_2^2)x + hy = 0$ seems to be not perpendicular to the major or transverse axis. For example see worked out Example No. 10.

Art. 51. (a) Working Rule for finding the Co-ordinates of foci, feet of directrices, vertices and equations of the latus rectum, directrices and tangents at the vertices.

On the line PQ, suppose A is a point whose coordinates (h, k) are known. The line PQ makes an angle θ with X axis. Then the co-ordinates of any ohter point (say) P (x, y) can be determined if the distance AP $(= d)$ is known. From the figure.

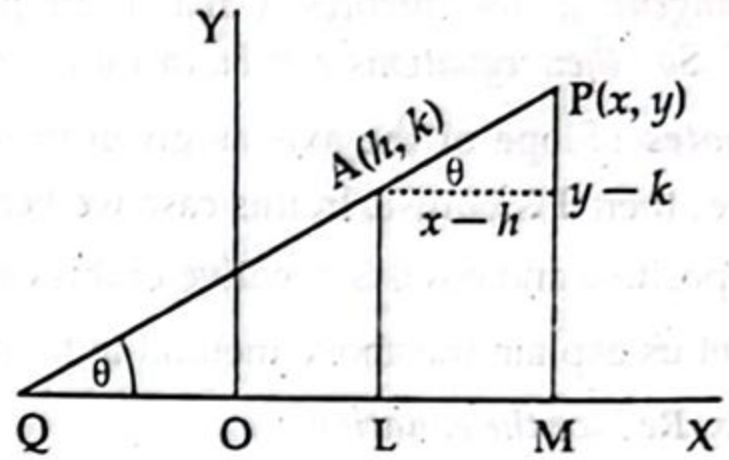


Fig : 20

$$\text{AP } (= d), \sin\theta = \frac{y-k}{\text{AP}} = \frac{y-k}{d} \text{ or, } y = k + d\sin\theta$$

$$\cos\theta = \frac{x-h}{\text{AP}} = \frac{x-h}{d} \text{ or, } x = h + d\cos\theta$$

Thus $x = h + d\cos\theta$

$y = k + d\sin\theta$ ... ... (XI)

Now let us try to apply in conic (Ellipse and Hyperbola)

Now let us try to apply in conic (Ellipse and Hyperbola)

The important points associated with them are the foci, vertices and feet of the directrices. These points lie either on the major axis or on the transverse axis. But centre of the conic is common to them. Therefore, we may take (h, k) as the co-ordinates of the centre of the conic and θ is the inclination of the major or transverse axis to the x axis. d stands for the distance between the centre and the required point on the conics. Now we are giving the distances of the points.

Distance for foci S and S′ ; $d = \pm ae = \pm r_1 e$

Distance of vertices A and A′; $d = \pm r_1$

Distance of the feet of the directrices Z and Z′ ; $d = \pm a/e = \pm r_1/e$ where r_1 is the length of semi major or semi transverse axis.

Thus d, h and θ are known, so any point S or S′ ; A or A′, Z or Z′ can be known.

Extremities of minor axis are B and B′. Here the distance d between the centre and B or B′ ; $d = \pm b = \pm r^2$; and θ slope of minor axis is known.

Length of the latus rectum LSL′ $= \frac{2b^2}{a} = \frac{2r_2^2}{r_1}$

For the determination of the extremities of latus rectum, we are to consider the known point as focus and (h, k) are the co-ordinates of the focus here. The distance between S and L or L′ is given by

$d = \pm b^2/a = \pm r_2^2/r_1$

The slope of the latus rectum is the same as the slope of minor axis. Therefore θ is known.

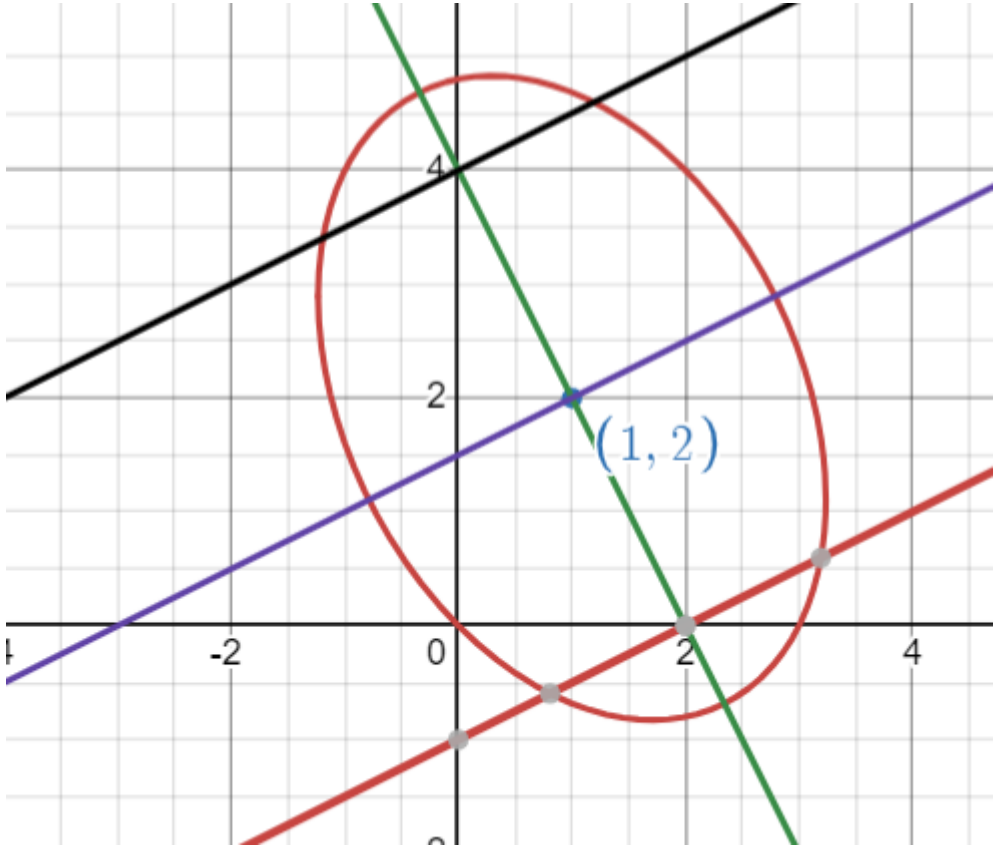
Expl: Reduce the equation

$$f(x,y) \equiv 8x^2 + 4xy + 5y^2 - 24x - 24y = 0 \ldots\ldots (i)$$

to its standard form and find all its properties.

Sol: Here $\Delta \neq 0$ and $ab - h^2 > 0$, so it's an ellipse.

- Find
- (i) Center
- (ii) Length of the axes
- (iii) Equation of major axis
- (iv) Equation of minor axis
- (v)Eccentricity
- (vi) Foci
- (vii) Vertices
- (viii) Length of the latus rectum
- (ix)Equation of the latus rectum
- (x) Equation of Directrices



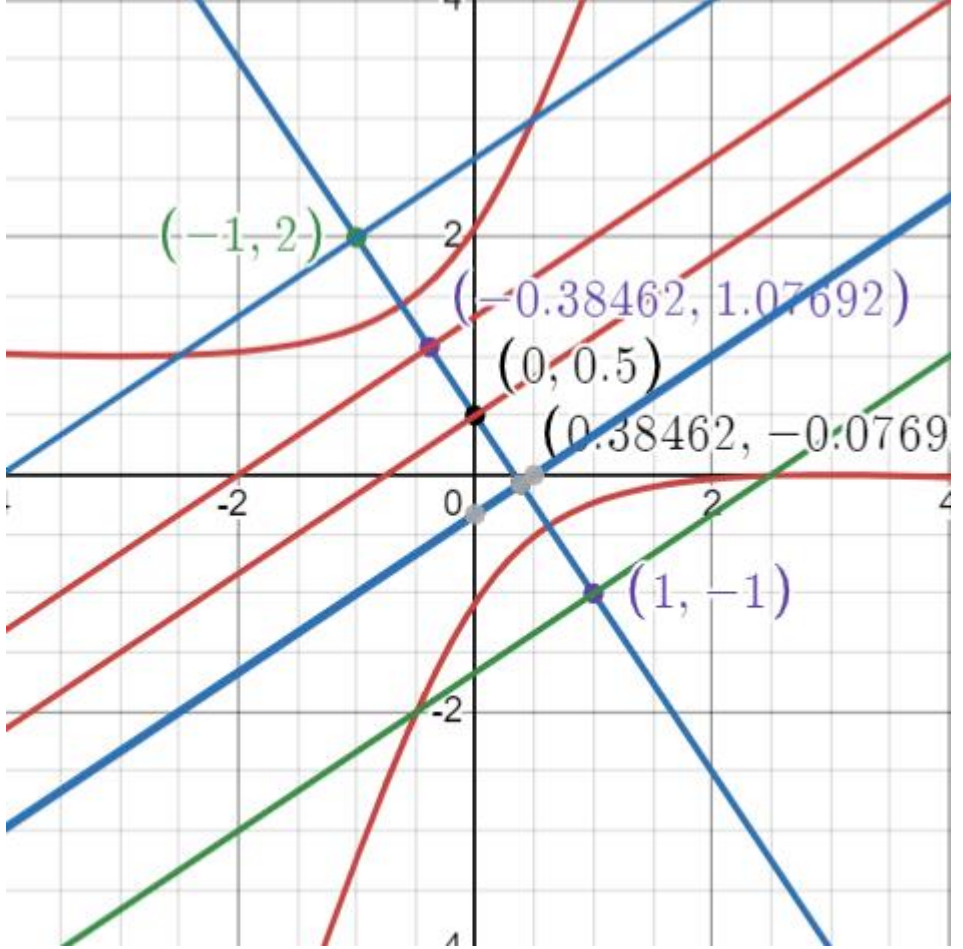
Expl: Reduce the equation

$$f(x,y) \equiv x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$$

to its standard form and find all its properties.

Sol: Solve as previous problem, and hence draw the graph using desmos.

- Find (i) Vertex
- (ii) Foci
- (iii) Equation of Directrices
- (iv)equation of transverse axis
- (v) Equation of conjugate axis
- (vi) Eccentricity
- (vii) Length of the latus rectum
- (viii) Length of the transverse axis
- (ix) Length of the conjugate axis



Expl. 14(g): Reduce the equation

$$f(x,y) \equiv x^2 + 2xy + y^2 + 2x - 1 = 0$$

to its standard form and find all its properties.

$$x^2 + 2xy + y^2 + 2x - 1 = 0$$

$$x + y + \frac{1}{2} = 0$$

$$-x + y + \frac{5}{4} = 0$$

- Find
- (i) Vertex
- (ii) Focus
- (iii) Equation of Directrix
- (iv) Length of the latus rectum

